

Jessi Pavia

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EXPERIENCE

Senior Software Engineer

April 2026 – Present

WMStudio

Remote

- Architected a 12-stage agentic video pipeline with headless orchestration, onboarding **200+ users** and reducing manual editing time by **60%**.
- Designed GCP distributed infrastructure with secure API auth and third-party integrations, sustaining **1.5k+ weekly renders** at **99.95% uptime**.

Founding Engineer

January 2026 – April 2026

aceAI

San Francisco, CA

- Architected **autonomous microservices** on GCP (Kubernetes) for high-throughput agentic workflows, driving **\$95k ARR** in 6 months while leading a team of 5.
- Engineered event-driven pipelines using **Pub/Sub**, processing **200k+ monthly events** at **99.9% uptime**.

FullStack Engineer Intern – Product Manager

January 2025 – December 2025

PlayAI (Acquired by Meta SuperIntelligence Labs)

Palo Alto, CA

- Built a real-time generative audio engine serving **1M+ daily active users** at **<200ms latency**.
- Implemented **CDN edge caching** across distributed regions, sustaining reliable performance for **70k+ monthly audio shares**.

Software Engineer

June 2023 – December 2024

Jinova

New York, NY

- Fine-tuned **Hugging Face** transformer models on proprietary datasets, reducing manual support workload by **50%** through specialized LLM deployment.
- Modernized data infrastructure with **Apache Beam**, accelerating ML deployment cycles by **40%**.

Software Engineer Intern – FullStack

February 2022 – May 2023

Trk Fndr

New York, NY

- Built a music discovery engine using **FAISS vector search**, delivering personalized recommendations to **2k+ users** via similarity-based retrieval.
- Implemented mixed-precision training pipelines in **PyTorch**, reducing cloud GPU costs by **\$5k/month** without degrading model accuracy.

PROJECTS

Voice AI Agent | *PlayAI, Node, Django, Next, JavaScript*

Jan 2025 – May 2025

- Built a distributed **autonomous voice agent**, with **3k+ daily interactions** with fault-tolerant error recovery
- Designed an AVL-tree text chunking algorithm optimizing **LLM** context window utilization, cutting memory usage by **40%** and response time by **200ms**.

ChatAPI (Enterprise RAG) | *NumPy, HuggingFace, Node, TypeScript*

Oct 2024 – Dec 2024

- Developed a context-aware **RAG system** with GPT-Neo (2.7B) and **tool calling**, serving **1k+ enterprise queries** at **95% accuracy**.
- Optimized retrieval pipelines via **Redis caching**, reducing deployment time by **30%** with multi-tenant support.

TECHNICAL SKILLS

Languages: Python (FastAPI), TypeScript, JavaScript, Go, Java, SQL

Backend: Django, Node.js, GraphQL, PostgreSQL, Redis, Celery, Pub/Sub

AI/ML: PyTorch, TensorFlow, Transformers (Hugging Face), LangChain, FAISS, RAG, LLM Ops, Agents

Infrastructure: AWS, GCP, Kubernetes, Docker, CI/CD, Terraform

Concepts: Autonomous Systems, Tool Calling, Event-Driven Architecture, Microservices, Distributed Systems

EXTRACURRICULAR ACTIVITIES

Hackathons: HackPrinceton, BobaHacks, MoonHack, Gemini API Developer, LifeHack (ML&AI), MLH

Olympiads: Putnam Competition, Math National Olympiads

EDUCATION

New Jersey Institute of Technology
Bachelor of Arts in Computer Science

Newark, NJ
May 2026